

Github: https://github.com/G-Kishor-IT/TCS_Training

```
1) import java.util.*;

public class Main {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();

        int[] arr = new int[n];

        int[] res = new int[n];

        int k = 0;

        for (int i = 0; i < n; i++) {

            arr[i] = sc.nextInt();

            if (arr[i] != 0) {

                res[k++] = arr[i];

            }

        }

        while (k < n) {

            res[k++] = 0;

        }

        for (int x : res) {

            System.out.print(x + " ");

        }

    }

}
```

```
2) import java.util.*;

public class Main {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();

        int bits = Integer.toBinaryString(n).length();

        int mask = (1 << bits) - 1;

        System.out.println(n ^ mask);

    }

}
```

```
3) import java.util.*;

public class Main {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        String day = sc.next().toLowerCase();

        int n = sc.nextInt();

        String[] days = {"sun","mon","tue","wed","thu","fri","sat"};

        int idx = 0;

        for(int i=0;i<7;i++){

            if(days[i].equals(day)){

                idx=i;

                break;

            }

        }

        int count = 0;

        for(int i=0;i<n;i++){

            if((idx+i)%7==0){

                count++;

            }

        }

        System.out.println(count);

    }

}
```

```
4) import java.util.*;

public class Main {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();

        int c0 = 0, c1 = 0, c2 = 0;

        for (int i = 0; i < n; i++) {

            int x = sc.nextInt();

            if (x == 0) c0++;

            else if (x == 1) c1++;

            else c2++;

        }

        while (c0-- > 0) System.out.print("0 ");

        while (c1-- > 0) System.out.print("1 ");

        while (c2-- > 0) System.out.print("2 ");

    }

}
```

```
5) import java.util.*;

public class Main {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();

        int max = Integer.MIN_VALUE;

        int count = 0;

        for (int i = 0; i < n; i++) {

            int x = sc.nextInt();

            if (x > max) {

                count++;

                max = x;

            }

        }

        System.out.println(count);

    }

}
```

```
6) import java.util.*;

public class Main {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        long n = sc.nextLong();

        long product = 1;

        while (n > 0) {

            product *= n % 10;

            n /= 10;

        }

        System.out.println(product);

    }

}
```

```
7) import java.util.*;

public class Main {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        String str = sc.next();

        int l = sc.nextInt();


        int max = 0;

        for (int i = 0; i < str.length(); i += l) {

            int count = 0;

            for (int j = i; j < Math.min(i + l, str.length()); j++) {

                if (str.charAt(j) == 'a') {

                    count++;

                }

            }

            max = Math.max(max, count);

        }

        System.out.println(max);

    }

}
```

```
8) import java.math.BigInteger;

import java.util.*;

public class Main {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();

        BigInteger fact = BigInteger.ONE;

        for (int i = 2; i <= n - 1; i++) {

            fact = fact.multiply(BigInteger.valueOf(i));

        }

        System.out.println(fact.multiply(BigInteger.valueOf(2)));

    }

}
```



```
9) import java.util.*;

public class Main {

    static int digitSum(long n) {
        int sum = 0;
        while (n > 0) {
            sum += n % 10;
            n /= 10;
        }
        return sum;
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        long n = sc.nextLong();
        int r = sc.nextInt();
        if (r == 0) {
            System.out.println(0);
            return;
        }
        int sum = digitSum(n) * r;
        while (sum >= 10) {
            sum = digitSum(sum);
        }
        System.out.println(sum);
    }
}
```